

An Internship Report
Of
Data Science Internship at
Cognifyz Technologies

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410030146

Name of Student: TUSHAR SINGH

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, TUSHAR SINGH, do hereby declare that the internship report titled "Data Science Internship at Cognifyz Technologies" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All references have been acknowledged, and I have not presented material from any other source as my own. I affirm that this report is based on my internship work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation in this regard.

Signature: 

Student Name: TUSHAR SINGH

Roll No.: 2410030146

ACKNOWLEDGEMENT

I take this opportunity to express my gratitude to everyone who supported me throughout the internship program. Their guidance, constructive feedback, and encouragement helped me complete the assigned tasks and improve my understanding of practical data analysis.

I sincerely thank Cognifyz Technologies for selecting me as a Data Science Intern and for providing an opportunity to work on a real restaurant dataset. The structured Level 1 and Level 2 tasks enabled me to practise data exploration, data cleaning, descriptive and geospatial analysis, service and price-range analysis, visualization, and feature engineering using Python.

I am also thankful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement. Finally, I express my gratitude to my parents for their blessings.

Signature: 

Student Name: TUSHAR SINGH

Roll No.: 2410030146

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3. INTERNSHIP COMPLETION CERTIFICATE



Cognifyz Technologies

Internship Completion Certificate

Date - 23/04/2026

This is to certify that **Tushar Singh, (Intern ID: CTI/A1/C328746)**, currently pursuing a B.Tech from The IILM UNIVERSITY, was working as a **Data Science Intern** with Cognifyz Technologies from 23/03/2026 to 23/04/2026.

During this period, he has served as a Data Science Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and effective communication abilities. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing his passion for Data Science. His commitment and willingness to acquire new knowledge and skills have been evident throughout his internship.

We extend our best wishes to Tushar Singh for a successful future, and we have no doubt that he will continue to excel in the field of Data Science.


With Regards,
Cognifyz Technologies



 cognifyztechnologies@gmail.com

 www.cognifyz.com

Internship Offer Letter



Nagpur, Maharashtra, India 📍
support@cognifyz.com ✉️
www.cognifyz.com 🌐

Date - 23/03/2026

INTERNSHIP OFFER LETTER

Dear Tushar Singh

Ref. : CTI/A1/C328746

Congratulations! We are pleased to inform you that you have been selected for the position of **Data Science Intern** at Cognifyz Technologies. We were impressed by your qualifications, skills, and enthusiasm for data analysis, and we are excited to have you join our team.

As a Data Science intern, you will have the opportunity to work on various projects and tasks, analyze complex datasets, and derive meaningful insights. Your role will involve collecting, cleaning, and interpreting data, as well as assisting in the development of data-driven solutions. We believe that your analytical mindset and attention to detail will contribute significantly to our organization's success.

We would like to take this opportunity to congratulate you once again and welcome you to Cognifyz Technologies. We are confident that this internship will provide you with valuable hands-on experience and further enhance your skills in the field of Data Science.

Best Regards,

Sukrit Chaudhary

Cognifyz Technologies



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work completed during the Data Science Internship at Cognifyz Technologies from 23 March 2026 to 23 April 2026. The internship was undertaken as part of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. Its central aim was to apply Python-based data analysis to a restaurant dataset and convert raw records into clear, evidence-based insights.

The source dataset contained 9,551 restaurant records and 21 original attributes. The attributes covered restaurant identity, country and city, address, geographic coordinates, cuisines, average cost for two, currency, table booking, online delivery, price range, aggregate rating, rating category, and votes. The completed workflow produced an enriched dataset with 27 columns after adding six engineered features.

4.2 Organization Profile

Cognifyz Technologies is presented in the supplied internship documents with the tagline "Where Data Meets Intelligence" and an office address in Nagpur, Maharashtra. Through the internship, the organization provided structured data science tasks focused on collecting, cleaning, interpreting, and visualizing data and deriving meaningful insights. The offer letter identifies the internship role as Data Science Intern under reference CTI/A1/C328746.

4.3 Problem Statement

Restaurant data combines numerical, categorical, and geospatial fields and may contain missing values, unrated entries, skewed numerical variables, and text-based service flags. The problem was to prepare this mixed dataset for analysis, describe its key distributions, identify cuisine and city patterns, examine whether location relates to rating, compare service availability and price ranges, and create machine-readable features for future modelling.

4.4 Project Objectives

- To inspect the restaurant dataset, verify its shape and data types, and identify missing values.
- To study the distribution and class imbalance of the Aggregate Rating target variable.
- To calculate descriptive statistics and identify leading cuisines and cities.
- To visualize restaurant locations and measure the relationship between coordinates and ratings.
- To compare table-booking and online-delivery availability and their relationship with ratings.
- To analyse restaurant counts and average ratings across the four price ranges.
- To engineer numerical features from text and Yes/No service fields and save an enriched dataset.
- To communicate the analysis through reproducible Python scripts, visualizations, and an HTML task report.

4.5 Scope of the Project

The project covers data loading, quality inspection, missing-value treatment, exploratory and descriptive analysis, univariate and grouped summaries, geospatial visualization, service and price-range comparisons, feature engineering, and export of an enriched CSV file. It uses the supplied restaurant dataset as the sole analytical source. The work is analytical and educational in scope: it does not include production deployment, a live recommendation system, causal inference, or a trained predictive model.

4.6 Technologies and Tools Used

- Programming language: Python.
- Data processing: pandas and NumPy.
- Visualization: Matplotlib and Seaborn.
- Development artifacts: level1_tasks.py and level2_tasks.py.
- Data files: Dataset .csv and Dataset_Enriched.csv.
- Reporting: HTML/CSS task report for presenting the verified statistics and insights.

4.7 System Architecture

The analysis followed a layered data-processing architecture. Each layer receives the output of the previous layer and produces a more useful representation of the restaurant data:

- Data Source Layer: reads the original 9,551-row CSV dataset.
- Preprocessing Layer: checks data types and fills nine missing Cuisine values with "Unknown".
- Exploratory Layer: summarizes ratings, numerical fields, cuisines, cities, costs, and votes.
- Geospatial Layer: filters zero coordinates, maps locations, and calculates coordinate-rating correlations.
- Business Analysis Layer: evaluates table booking, online delivery, and price-range behaviour.
- Feature Engineering Layer: creates six text-derived and binary service variables.
- Output Layer: saves the 27-column enriched dataset, figures, and task report.

4.8 Methodology and Task Execution

The internship tasks were completed in two progressive levels. Level 1 established the quality and descriptive profile of the dataset; Level 2 examined business features and created model-ready variables. The six-task workflow is summarized below.

Task	Module	Work Completed
L1-T1	Data Exploration and Preprocessing	Loaded the CSV, checked 9,551 x 21 shape, inspected types and missing values, filled nine missing cuisine entries, and analysed rating imbalance.
L1-T2	Descriptive Analysis	Calculated numerical summaries and identified top cuisines, cities, cost patterns, and vote distributions.
L1-T3	Geospatial Analysis	Removed zero coordinates, mapped valid locations, produced an India-region view, and measured coordinate-rating correlations.
L2-T1	Table Booking and Online Delivery	Calculated availability percentages, compared average ratings by booking status, and measured online delivery within each price range.
L2-T2	Price Range Analysis	Found the most common price range, restaurant counts, average ratings, and the rating colour associated with the highest-rated range.
L2-T3	Feature Engineering	Created six numerical features, measured their correlations with rating, and exported Dataset_Enriched.csv with 27 columns.

The transformed output was saved separately as Dataset_Enriched.csv.

4.8.1 Level 1 Results: Exploration, Description, and Location

The initial inspection found only nine missing values, all in the Cuisines column. After replacement with "Unknown", no missing values remained. Aggregate Rating ranged from 0.0 to 4.9, with a mean of 2.67, median of 3.20, and standard deviation of 1.52. A total of 2,148 restaurants (22.49%) had a rating of 0.0, indicating a substantial unrated class that should be considered before predictive modelling.

North Indian was the most frequent cuisine (3,960 occurrences), followed by Chinese (2,735), Fast Food (1,986), Mughlai (995), and Italian (764). New Delhi contained 5,473 restaurants, followed by Gurgaon (1,118), Noida (1,080), and Faridabad (251). Average Cost for Two was strongly right-skewed: its mean was 1,199.21 while its median was 400. Votes showed a similar concentration, with a mean of 156.91, a median of 31, and a maximum of 10,934.

After excluding records with zero latitude or longitude, 9,052 restaurants had valid coordinates; 8,155 fell within the India-region bounding box used in the analysis. Latitude-rating correlation was -0.124 and longitude-rating correlation was -0.206. These weak linear relationships indicate that coordinates alone do not adequately explain restaurant rating.

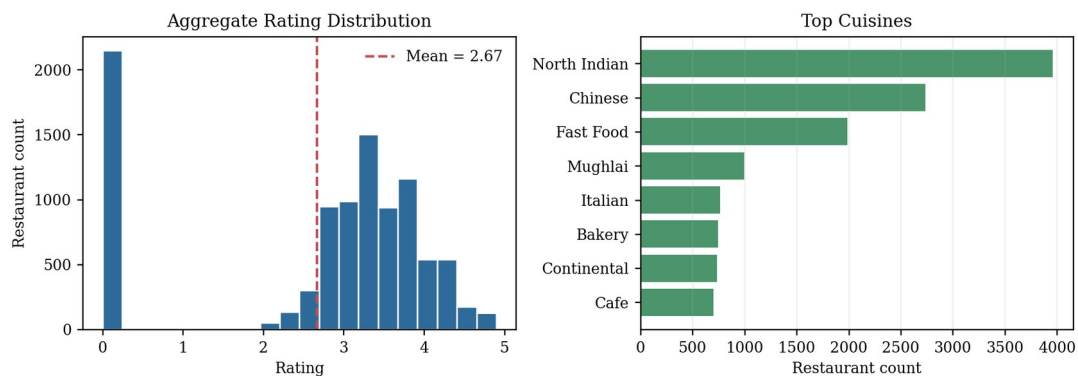


Figure 1. Rating distribution and the most frequent cuisines in the dataset.

4.8.2 Level 2 Results: Services and Price Range

Only 12.12% of restaurants offered table booking, while 25.66% offered online delivery. Restaurants with table booking had an average rating of 3.442 compared with 2.559 for restaurants without table booking, a descriptive difference of 0.883 rating points. This comparison shows association, not causation.

Online delivery was most common in Price Range 2 (41.31%), followed by Price Range 3 (29.19%), Price Range 1 (15.77%), and Price Range 4 (9.04%). Price Range 1 was the largest category with 4,444 restaurants, followed by Price Range 2 with 3,113, Price Range 3 with 1,408, and Price Range 4 with 586.

Average rating increased consistently with price range: 2.00 for Range 1, 2.94 for Range 2, 3.68 for Range 3, and 3.82 for Range 4. Price Range 4 therefore had the highest average rating, and Yellow was the most common rating colour associated with that group.



Figure 2. Service availability and average aggregate rating across price ranges.

4.8.3 Feature Engineering

Six numerical features were derived to make textual and service information easier to analyse and use in a future model. The enriched dataset contains all 21 original fields plus these six variables, for a final shape of 9,551 rows by 27 columns.

Feature	Definition	Correlation with Rating
Restaurant Name Length	Number of characters in the restaurant name	-0.035
Address Length	Number of characters in the address	+0.002
Cuisine Count	Number of cuisine labels listed	+0.249
Has Table Booking Enc	1 for Yes; 0 for No	+0.190
Has Online Delivery Enc	1 for Yes; 0 for No	+0.226
Is Delivering Now Enc	1 for Yes; 0 for No	+0.019

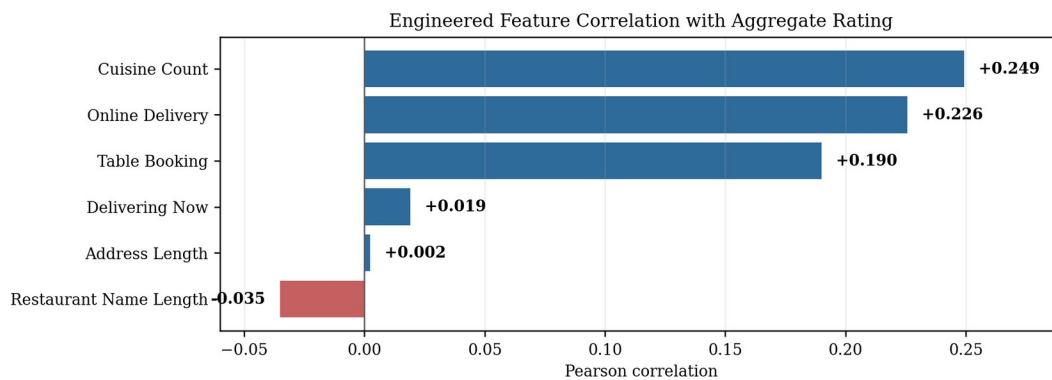


Figure 3. Correlation of engineered features with Aggregate Rating.

4.9 Outcomes and Key Findings

Cuisine Count was the strongest engineered predictor ($r = +0.249$), followed by online delivery ($r = +0.226$) and table booking ($r = +0.190$). Name length, address length, and current-delivery status had negligible linear relationships with rating. The internship therefore produced a clean, documented, and enriched dataset together with a reproducible exploratory workflow and interpretable business findings.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate and Internship Offer Letter issued by Cognifyz Technologies are attached in Chapter 3. Together they confirm Tushar Singh's selection and completion of the Data Science Internship under Intern ID CTI/A1/C328746 for the period 23 March 2026 to 23 April 2026.

The following supplied work products provide supporting evidence of the tasks completed during the internship:

Artifact	Purpose / Evidence
Dataset .csv	Original restaurant dataset containing 9,551 rows and 21 columns.
level1_tasks.py	Data exploration, preprocessing, descriptive analysis, and geospatial analysis.
level2_tasks.py	Booking/delivery analysis, price-range analysis, and feature engineering.
Dataset_Enriched.csv	Final dataset containing the original fields plus six engineered features.
Cognifyz_Task_Report.html	Formatted summary of Level 1 and Level 2 task results and insights.

The quantitative statements in this report were cross-checked against the original CSV file and the logic implemented in both Python scripts. The enriched file was also checked to confirm its 9,551 x 27 structure and the presence of all six new features.

5. BIBLIOGRAPHY/REFERENCES

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